

GENERICSGENERICS

```
// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }
// Full Example
class OrderedPair<T, U> {
    T first;
    U second;
    public OrderedPair(T t, U u) {
        this.first = t;
        this.second = s;
    }
    public T getFirst() { return first; }
    public U getSecond() { return second; }
    @Override
    public int hashCode() {
        return Objects.hash(first.hashCode(),
            second.hashCode());
    }
    @Override
    public boolean equals(Object obj) {
        if (obj == null ||
            obj.getClass() != getClass()) return false;
        @SuppressWarnings("unchecked")
        var other = (OrderedPair<T, U>)obj;
        return Objects.equals(first, other.first) &&
            Objects.equals(second, other.second);
    }
}

public class Employee implements Comparable<Employee> {
    // ...
    @Override
    public int compareTo(Employee other) {
        return Double.compare(this.salary, other.salary);
    }
}
```

JAVA BEING STUPID

```
T obj = new T(); // error
obj instanceof T; // error
T heh = (T) "asdf"; // no typechecking
var s = Stack<int>(); // no primitive types
private static T x; // static no workey
Node<Number> s = new Node<Integer>; // error

class IThrowAnErrorBecauseWhyNot {
    void m(List<String> list) { }
    void m(List<Integer> list) { }
}
// error because of type erasure & identifiability
static <T extends Comparable<T>> T m(T x, T y, T z) {
    return x > y ? z : x;
}

Integer n = m(1, 3.141, 0); // error
Number n = m(1, 3.141, 0); // ok
```

COMPARABLE

```
interface Comparable<T> {
    int compareTo(T o);
}

static <T extends Comparable> T doStuff(T a, T b) {
    // compareTo is possible with all types
    return a.compareTo("b") > 0 ? a : b;
}

static <T extends Comparable<T>> T doStuff(T a,T b) {
    // compareTo is possible only with T
    return a.compareTo(b) > 0 ? a : b;
}
```

EXAMPLE

```
class Graphic {
    public void draw() { }
}

class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
    extends Stack<T> {
    public void drawAll() {
        for (T item : this) item.draw();
    }
}
```

BYTECODE

JVM Architecture:

- Operand Stack (LIFO op vals)
- Local variables (op results)
- Constant pool (runtime refs)

TYPE ERASURE

– Introduced because of "bAcKwArDs CoMpAtAbIlItY"

– No type information at runtime (Non-Reifiable Type)

```
class MyStack<T> {
    Entry<T> top;
    int push(T value) { }
}
// becomes
class MyStack {
    Entry top;
    int push(Object value) { }
}
/* with bounds */
class MyStack<T extends Comparable<T>> {
    Entry<T> top;
    int push(T value) { }
}
// becomes
class MyStack {
    Entry top;
    int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();
```

WILDCARDS

```
void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1); // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2); // error
/* solution */
void printGs(List<? extends Graphic> gs) { }
```

GENERIC VARIANCE

	Type	Compatible Type-Args	R	W
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	X
Contravariance	C<? super T>	T and basis-types	X	✓
Bivariance	C<?>	All	X	✓

INVARIANCE

```
static <T> void move(Stack<T>, from, Stack<T> to) {
    while(!from.isEmpty()) to.push(from.pop());
}

var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle()); // ok
move(rectS, graphS); // error
Stack<Object> objS = graphS; // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA; // ok
objA[0] = Integer.valueOf(2); // runtime err
/* anotherone */
ArrayList<String> sA = new ArrayList<>();
ArrayList<Object> oA = sA; // error
```

TODO:

```
// compiles?
List<Graphic> list = new ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
```

```
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
for (Rectangle r : rectangleStack) {
    graphicStack.push(r);
}
```

COVARIANCE

```
public class Stack<E> {
    public void pushAll(Iterable<? extends E> src) { }
}

Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>(); // ok
graphS = new Stack<Circle>(); // ok
graphS = new Stack<Object>(); // error (duh)
/* read only */
graphS.push(new Graphic()); // error
graphS.push(new Rectangle()); // error
graphS.push(new Triangle()); // error
```

CONTRAVARIANCE

```
static <T> void addToC(List<? super T> list, T e) {
    list.add(e);
}

addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok
/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic()); // ok
s.add(new Rectangle()); // ok
Graphic g = s.pop(); // error
Object g = s.pop(); // ok
```

BIVARIANCE

– If concrete type doesn't matter

```
static void printList(List<?> list) {
    for (Object elem : list)
        System.out.print(elem + " ");
}

/* more shenanigans */
static void appendNewObject(List<?> list) {
    list.add(new Object()); // error
}
```

ANOTHERONE

```
public class VarianceExamples {
    public static double sum(
        Collection<? extends Number> numbers) {
        double sum = 0;
        for (Number num : numbers)
            sum += num.doubleValue();
        return sum;
    }

    public static void addNumbers(List<? super Integer>
        list, Collection<? extends Integer> source) {
        list.addAll(source);
    }

    public static <T extends Comparable<T>> T
        findMax(Collection<T> coll) {
        return coll.stream()
            .max(Comparator.naturalOrder())
            .orElse(null);
        }

    public static <T> List<T> filterByType(
        Collection<?> source, Class<T> clazz) {
        List<T> result = new ArrayList<>();
        for (Object obj : source)
            if (clazz.isInstance(obj))
                result.add(clazz.cast(obj));
        return result;
    }
}
```

PRODUCER / CONSUMER

from **produces** entries, to **consumes** entries

```
<T> void move(Stack<? extends T> from,
    Stack<? super T> to) {
    while (!from.isEmpty())
        to.push(from.pop());
}
```

MISC

<T extends Comparable & Collection> // multiple type bounds

```
Set<Integer> set = new Set<>();
Integer[] c = set.toArray(new Integer[0]);

RECORDS
public record Person (String name, String address) {}

GENERICS STREAM TOMFOOLERY
List<? extends Media> mediaList;
public <S extends T> List<S>
    filterBySubtype(Class<S> subtype) {
        return mediaList.stream()
            .filter(subtype::isInstance)
            .map(subtype::cast)
            .collect(Collectors.toList());
    }
}
```

ANNOTATIONS

- Metadata
- Not actually part of the code

TODO:

slides 10

- @Override
- @Test
- @Json...

DEFINING

```
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
    String name();
    String date() default "N/A";
}

@Author(name = "UwU", date = "idon't validate input")
public class Wee0oo { }
```

TODO:

Examples Woche03

REFLECTION

- Information of Class (private and public): Methods, Attributes, Parent class, Implemented interfaces
- Calling methods possible
- Usages: Debugger, Serialization, Frameworks, Remote Procedure Call, ORM

// all of these `throws SecurityException`

```
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<>[] getDeclaredConstructors()

System.out.println(dump(new Student("a", "b"), 0));
// {
//     firstName = a
//     lastName = b
// }

Class c = "s".getClass(); // java.lang.String
```

METHOD

```
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD)
public @interface Init { }
// ...

public class User {
    private String needsInit;
    @Init
    private void initObj() {
        this.needsInit = "wowie";
    }
}
```

CLASS

TODO:

```
getDeclaredConstructor().newInstance()

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.TYPE)
public @interface JsonSerializable { }
```

```
// ...
@JsonSerializable
public class User { }

ATTRIBUTE
@Retention(RetentionPolicy.RUNTIME)
@Target({ ElementType.FIELD })
public @interface JsonElement {
    public String key() default "";
}

// ...
public class User {
    @JsonElement
    private String firstName;
}
```

VALIDATION

```
@Min(value = 18, message = "Age must be ≥ 18")
@Max(value = 99, message = "Age must be ≤ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;

EXAMPLE
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface WebController { }
// ...
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD)
public @interface Get {
    String path();
    String contentType() default "application/json";
}

// ...
@WebController
public class Controller {
    private User user = new User("frank", "meier");
    @Get(path = "/user")
    public String user() {
        returnn new ObjectToJsonConverter()
            .convertToJson(user);
    }
}
```

```
// ...
public class WebFramework {
    public void addController(Class<?> controllerClass)
        throws Exception {
        if (!controllerClass
            .isAnnotationPresent(WebController.class))
            throw new Exception("Is not a WebController");

        for (Method method : controllerClass
            .getDeclaredMethods())
            if (method.isAnnotationPresent(Get.class)) {
                method.setAccessible(true); // best practice or
                sth, idk, i write code in actually good languages
                var annot = method.getAnnotation(Get.class);
                routeHandlers.put(
                    annot.path(),
                    new WebHandler<>(method,
                        controllerClass
                            .getDeclaredConstructor()
                                .newInstance(),
                                annot.contentType()));
            }
}
```

ALGORITHMS

Brute-force: slow, stupid, go shame yourself

Greedy: Take best option each step (optimize locally). Fast. Not necessarily global optimum

Backtracking: Trial and error. Best overall option. Higher compute costs

Divide and conquer: Binary search

Dynamic programming: Recursion optimization? Tabulation. Iterative reuse of previous results

BACKTRACKING

TODO:

slides 87

TODO:
example

KNIGHTSTOUR

TODO:
diagram

```
boolean tour(int[][] visited, int x,int y, int pos) {
    visited[x][y] = pos;
    if (pos >= N * N) return true;
    for (int k = 0; k < 8; k++) {
        int newX = x + row[k];
        int newY = y + col[k];
        if (isValid(newX, newY)
            && visited[newX][newY] == 0
            && tour(visited, newX, newY, pos + 1))
            return true;
    }
    visited[x][y] = 0;
    return false;
}
```

BIG O NOTATION

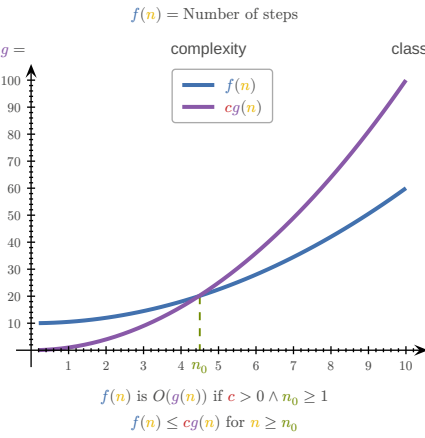
TODO:
tricky examples

- Upper bound of f
- Worst case scenario
- Atomic operations = constant time
- Runtime measured as sum of primitive operations

Notation

f	Algorithm	n	Size of Problem
g	Complexity Class	c	> 0

$n_0 \geq 1$



f in order g :

$f, g : \mathbb{N} \rightarrow \mathbb{N}$
 $n_0, c \in \mathbb{N} \wedge \forall n \geq n_0. (f(n) \leq c \cdot g(n))$
 $\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$
 $O(n)$ = Complexity class

RULES

If $f(n)$ is polynomial of degree d , then $f(n) \in O(n^d)$

- ignore lower powers
- ignore constants

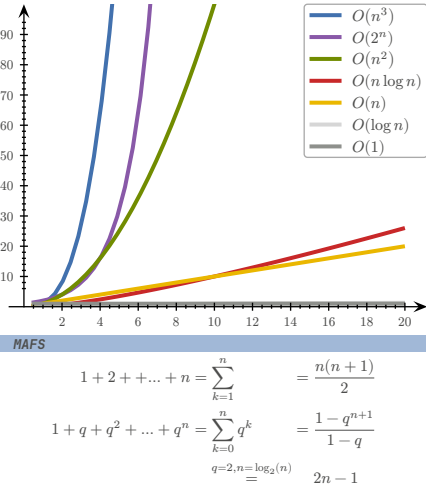
Use most optimal function (lowest power)

- $2n$ is $O(n)$ better than $2n$ is $O(n^2)$

Simplify as far as possible

- $3n + 5$ is $O(n)$ instead of $3n + 5$ is $O(3n)$

COMPLEXITY CLASSES		
Name	Class	Example
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations



SORTING ALGORITHMS

TODO:
Sorting diagrams

BOGOSORT

The best sorting algorithm

STALINSORT

The most efficient sorting algorithm

SELECTION SORT

- Search for smallest elem in unsorted part and move in front
- Less mutations in array
- Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2 - n}{2} \sim O(n^2)$$

INSERTION SORT

- Take elem and put into correct place in sorted part
- Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
    int n = a.length;
    for (int i = 1; i < n; i++)
        for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
            swap(a, j, j - 1);
}
```

$$1 + \dots + (n-1) + n = \sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2 + n}{2} \sim O(n^2)$$

BUBBLE SORT

- Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
    for (n = a.length; n > 1; n--)
        for (i = 0; i < n - 1; i++)
            if (a[i] > a[i + 1])
                swap(a, i, i + 1);
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2 - n}{2} \sim O(n^2)$$

COUNTING SORT

```
public static void countingSort(int[] a) {
    int k = Arrays.stream(arr).max().getAsInt();
    int[] c = new int[k + 1];
    for (int i : a) c[i] += 1;
    int pos = 0;
    for (int i = 0; i < k; i++)
        for (int j = 0; j < c[i]; j++) {
            a[pos] = i;
            pos++;
        }
}
```

$$n + n + k + n = 3n + k \sim O(n + k)$$

SHELL SORT

- Sort pairs of far away elems
- Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
    int n = a.length;
    int h = 1;
    while (h < n/3) h = 3 * h + 1;
    while (h >= 1) {
        for (int i = h; i < n; i++)
            for (int j = i; j >= h && a[j] < a[j - h]; j = j - h) swap(a, j, j - h);
        h /= 3;
    }
}
```

BINARY SEARCH

TODO:
iterative vs recursive (Woche04 Aufgabe3)

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
    if (low > high) return false;
    int pivot = low + ((high - low) / 2);
    if (target.equals(data.get(pivot)))
        return true;
    else if (target.compareTo(data.get(pivot)) < 0)
        return searchBinary(data, target, low, pivot - 1);
    else
        return searchBinary(data, target, pivot + 1, high);
}
```

$$O(\log n)$$

RECURSION

When a function calls itself. FP <3

LINEAR RECURSION

Recursive call starts **at most one** further recursive call

RECURSIVE $\rightarrow$ ITERATIVE

```
static int[] reverseArrRec(int[] a, int i, int j) {
    if (i < j) {
        int temp = a[j];
        a[j] = a[i];
        a[i] = temp;
        reverseArray(a, i + 1, j - 1);
    }
    return a;
}
```

```
static int[] reverseArrIter(int[] a, int i, int j) {
    while (i < j) {
        int temp = a[j];
        a[j] = a[i];
        a[i] = temp;
        i = i + 1;
        j = j - 1;
    }
}
```

```
return a;
}
```

TAIL RECURSION

Linear recursion where the recursive call is **last**

```
static int recsum(int x) {
    if (x == 0) return 0;
    else return x + recsum(x - 1);
    // no tail recursion because "+" is evaluated last
}
```

```
static int tailrecsum(int x, int total) {
    if (x == 0) return total;
    else return tailrecsum(x - 1, total + x);
}
```

BINARY RECURSION

All non-terminal calls have **two** recursive calls

```
static int binsum(int low, int high) {
    if (low > high)
        return 0;
    else if (low == high)
        return low;
    else {
        int mid = (low + high) / 2;
        return binsum(low, mid) + binsum(mid + 1, high);
    }
}
```

DATA STRUCTURES

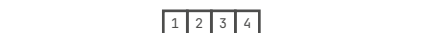
TODO:
code examples

ADT: Abstract Data Type (e.g. Interface). "What", not "How"



ARRAY

TODO:
data structures $O(n)$ operations



LINKED LIST



DOUBLY LINKED LIST



STACK



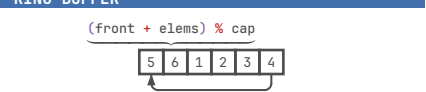
QUEUE



FIFO



TREE



```
class RingBuffer {
    private int[] buffer;
    private int size;
    private int start;
    private int end;
    private boolean isFull;

    public RingBuffer(int size) {
        if (size <= 0)
            throw new IllegalArgumentException("Size must be greater than 0");
        this.size = size;
        this.buffer = new int[size];
    }
}
```

TODO:

Woche09 Aufgabe4

```
this.start = 0;
this.end = 0;
this.isFull = false;
}
```

```
public void append(int item) {
    buffer[end] = item;
    end = (end + 1) % size;
    if (isFull)
        start = (start + 1) % size;
    if (end == start)
        isFull = true;
}
```

```
public List<Integer> getData() {
    List<Integer> items = new ArrayList<>();
    if (!isFull) {
        for (int i = 0; i < end; i++)
            items.add(buffer[i]);
    } else {
        for (int i = start; i < size; i++)
            items.add(buffer[i]);
        for (int i = 0; i < end; i++)
            items.add(buffer[i]);
    }
    return items;
}
```

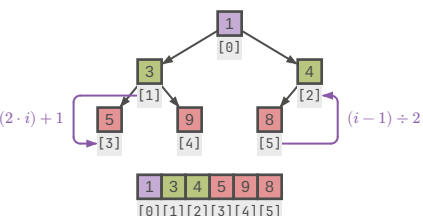
PRIORITY QUEUE		
Method	Unsorted List	Sorted List
insert	$O(1)$	$O(n)$
min	$O(n)$	$O(1)$
removeMin	$O(n)$	$O(1)$

HEAP

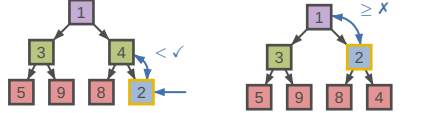
- Binary tree
- Layers $0, \dots, h-1$ fully populated, layer h filled from left
- Operations always lowest RHS so that tree stays consistent

Min-Heap: Keys of nodes are **smaller** or equal than children's

Max-Heap: Keys of nodes are **larger** or equal than children's



ENQUEUE



DEQUEUE



HEAP SORT

Dequeue elems from head into list, resorting tree on each iter

TODO:

TODO:
Woche09 Aufgabe4

BINARY SEARCH TREE (BST)

- All subnodes of **left** subtree are **smaller** than root
- All subnodes of **right** subtree are **larger** than root

